THE NUMBER OF GEOMETRIC CONNECTED COMPONENTS OF FIBRES OF A PROPER FLAT FINITELY PRESENTED MAP IS LOWER SEMI-CONTINUOUS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8349-8358 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8360-8384, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (both complete); depends on L16200 (same wave) and one queue card (wave 21).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter theorems/lemmas stein-factorization-general (L16200), stein-universally-closed-residue-fields, base-change-fibres-nr-geometrically-connected-components; More on Morphisms, lemma etale-makes-integral-split; Morphisms of Spaces, lemma fppf-open.

Source reference. Original lemma 055F: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7070–7084. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E0M: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8061–8066. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BSR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8384–8384. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 042S: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8384–8384. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03H2: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8197–8214. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $X \to Y$ be a morphism of algebraic spaces over S. If f is proper, flat, and of finite presentation, then the function $n_{X/Y} : |Y| \to \mathbf{Z}$ counting the number of geometric connected components of fibres of f (Lemma 055F) is lower semi-continuous.*

Proof. The question is étale local on Y, hence we may and do assume Y is an affine scheme. Let $y \in Y$. Set $n = n_{X/S}(y)$. Note that $n < \infty$ as the geometric fibre of $X \to Y$ at y is a proper algebraic space over a field, hence Noetherian, hence has a finite number of connected components. We have to find an open neighbourhood V of y such that $n_{X/S}|_V \geq n$. Let $X \to Y' \to Y$ be the Stein factorization as in Theorem 03H2. By Lemma 0E0M there are finitely many points $y'_1, \ldots, y'_m \in Y'$ lying over y and the extensions $\kappa(y'_i)/\kappa(y)$ are finite. More on Morphisms, Lemma 0BSR tells us that after replacing Y by an étale neighbourhood of y we may assume $Y' = V_1 \amalg \ldots \amalg V_m$ as a scheme with $y'_i \in V_i$ and $\kappa(y'_i)/\kappa(y)$ purely inseparable. Then the algebraic spaces $X_{y'_i}$ are geometrically connected over $\kappa(y)$, hence $m = n$. The algebraic spaces $X_i = (f')^{-1}(V_i)$, $i = 1, \ldots, n$ are flat and of finite presentation over Y. Hence the image of $X_i \to Y$ is open (Morphisms of Spaces, Lemma 042S). Thus in a neighbourhood of y we see that $n_{X/Y}$ is at least n. □